Sumit Singh Bagdwal

Target: Software Engineering Intern — Summer 2027

Uttarakhand, India | +91 9105547410 | sumitsbagdwal@gmail.com
Portfolio | LinkedIn | GitHub | LeetCode

Education

Graphic Era Hill University — Bhimtal, Uttarakhand
B.Tech in Computer Science and Engineering | CGPA: 8.26/10 — *2024 – Expected 2028*

Technical Skills

Languages: C++, JavaScript (ES6+), Python, SQL
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC, WebSockets, Socket.IO
Databases & Cloud: MongoDB Atlas, Mongoose, MySQL, Redis, AWS EC2/S3/IAM, Docker, Vercel
Frontend: React.js, Vite, HTML, CSS, Tailwind CSS
Development & Testing: Git, GitHub, GitHub Actions, Automated Testing, CI/CD, Postman, GitHub Copilot, Claude Code
Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, SDLC, Agile/Scrum, Distributed Systems

Projects

IBVAP – Intelligent Border Video Analytics | *FastAPI, Node.js, YOLO11, AWS* — GitHub

- Developed a real-time surveillance prototype **shortlisted for Smart India Hackathon 2026**, combining YOLO11/ByteTrack tracking with configurable risk rules and **alert deduplication**.
- Deployed a Dockerized Node.js/Express backend on **AWS EC2**; connected FastAPI edge inference over Tailscale with MySQL, Redis, and real-time event handling.
- Implemented **SHA-256/Ed25519 evidence verification**, RBAC, and camera reconnection; documented **407 passing AI tests and 41 frontend tests** in local validation.

Pratyaya – Payment Reliability Platform | *Node.js, React, MongoDB Atlas* — GitHub

- Built a simulated, merchant-scoped **payment platform** with Express REST APIs, JWT/RBAC, and a React interface for credential-free, role-based sandbox exploration.
- Developed payment **idempotency controls**, double-entry ledger workflows, and a transactional outbox using **MongoDB ACID transactions** for auditable financial records.
- Integrated mock-provider operations, MongoDB-backed jobs, and reconciliation workflows; recorded **42 passing API and 14 provider integration tests** against MongoDB Atlas.

Professional Experience

Lead, AWS Student Builder Group | *Student Volunteer, GEHU Bhimtal* — 2026 – Present

- Lead collaborative student learning sessions in software development, cloud computing, and problem solving; helped organize the **INNOVATE 1.0** campus ideathon.

Achievements & Training

- **Hackathons & DSA:** Smart India Hackathon 2026 shortlist; Hack the Winter 2025 national finalist; **350+ LeetCode problems** solved.
- **NPTEL:** Data Structures and Design.
- **AWS:** Cloud Essentials and Cloud Practitioner Essentials training.
- **NCC:** National Cadet Corps B Certificate.